Macroeconomics I — MPE Insper

Practice list: The New Keynesian AS–AD Model — graduate level (Lista 7)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Notation (Benigno 2015). Lower-case letters are log deviations from the steady state; bars denote the long run. Households: $u(C) - v(L) + \beta\{u(\bar{C}) - v(\bar{L})\}$, $u = C^{1-\tilde{\sigma}^{-1}}/(1-\tilde{\sigma}^{-1})$, $v = L^{1+\eta}/(1+\eta)$, $\rho \equiv -\ln\beta$. Resources: $y = s_c c + g$, $\sigma \equiv \tilde{\sigma} s_c$. Firms: demand $Y(j) = (P(j)/P)^{-\theta}(C+G)$, technology $Y(j) = AL(j)$; a fraction α of prices is preset at P^e. Calibration used throughout unless stated: $\alpha = 0.75$, $\sigma = 0.5$, $\eta = 0.2$, $\theta = 8$, $\rho = 0.01$ (a quarter), $p^e = \bar{p} = 0$, $g = \bar{g} = 0$, no taxes.

Block 1 — Foundations: AD, AS, natural and efficient output

(Benigno §§1–5 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) The slope κ of the AS curve $p - p^e = \kappa(y - y_n)$ under the calibration is:

A. 6.600

B. 0.733

C. 0.233

D. 2.200

b) (3 points) Productivity rises by $a = 0.02$ and the markup by $\mu = 0.05$, both temporarily. The changes in $(y_n,\ y_e)$ are:

A. $(-0.0227,\ 0)$

B. $(0.0109,\ 0.0109)$

C. $(-0.0118,\ -0.0118)$

D. $(-0.0118,\ 0.0109)$

c) (2 points, T or F) Because θ sets the markup, a higher demand elasticity θ makes the AS curve steeper.

d) (2 points, T or F) Holding i, $\bar{p}$ and p^e fixed, news that long-run government spending will be higher ($\bar{g}\uparrow$, g unchanged) lowers current output.

Question 2. (15 points) Discursive. Build the model from its primitives (no taxes, τ's $= 0$).

a) (3 points) From the household problem with budget $PC + \frac{\bar{P}\bar{C}}{1+i} = WL + \frac{\bar{W}\bar{L}}{1+i} + T$, derive the Euler equation and the labor-supply condition. Log-linearize the Euler equation and combine it with $y = s_c c + g$ to obtain the AD curve. Give its slope in the (y, p) plane and its shifters.

Solution:

b) (4 points) From the firm's problem derive the optimal flexible price, then the relation

$\tilde{p} - p = (\sigma^{-1} + \eta)(y - y_n)$ and y_n. With a fraction α of prices preset at p^e (and $p = \alpha p^e + (1-\alpha)\tilde{p}$ to first order), derive the AS curve and κ.

Solution:

c) (4 points) Derive efficient output y_e from the planner's problem and prove $y_n - y_e = -\mu/(\sigma^{-1} + \eta)$. Prove that κ does not depend on θ. Give the limits of the AS as $\alpha \to 0$ and $\alpha \to 1$. What subsidy makes the steady state efficient?

Solution:

d) (4 points) An oil shock raises the markup by $\mu = 0.05$ while productivity falls by $a = -0.02$, both for the short run only. Prove that the equilibrium gap is $y - y_n = \frac{\sigma}{1+\sigma\kappa}[r_n - i + (\bar{p} - p^e)]$ with $r_n = \rho + \sigma^{-1}(\bar{y}_n - y_n)$. Compute y_n, y_e, r_n; the outcome if the bank keeps $i = \rho$; and the i that delivers $p = p^e$. Where is output relative to y_e in each case?

Solution:

Block 2 — Policy: optimal rules, the zero bound and deleveraging

(Benigno §§6–12 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) A temporary cut in consumption taxes of 2 points (like the 2009 IPI cut on cars, modeled as $\tau_c = -0.02$, entering the markup one for one) changes the natural real rate $r_n = \rho + \sigma^{-1}(\bar{y}_n - y_n) + \sigma^{-1}(g - \bar{g}) + (\bar{\tau}_c - \tau_c)$ by:

A. $+2.00$ points

B. -1.82 points

C. $+0.18$ points

D. 0

b) (3 points) With i fixed, a temporary rise in g of 1% of GDP changes output and the gap $y - y_n$ by, respectively:

A. 1.000% and 0.000%

B. 0.909% and 0.000%

C. 0.732% and 0.067%

D. 0.976% and 0.067%

c) (2 points, T or F) At the zero lower bound with $r_n < 0$, a credible commitment to a long-run price level higher by $-r_n$ restores $y = y_n$ and $p = p^e$.

d) (2 points, T or F) In the borrower–saver economy with constant inflation between the short and the long run, more price flexibility (higher κ) reduces the output cost of a deleveraging shock.

Question 4. (15 points) Discursive. Optimal policy, the zero bound and deleveraging.

a) (3 points) The bank minimizes $L = \frac{1}{2}(y - y_e)^2 + \frac{1}{2}\frac{\theta}{\kappa}(p - p^e)^2$ (eq. 33) subject to the AS. Derive the targeting rule $(y - y_e) + \theta(p - p^e) = 0$, describe it in the (y, p) plane and prove that shocks that leave μ unchanged create no trade-off.

Solution:

b) (4 points) For the shock of Question 2(d), compute the optimal y, $p - p^e$ and i and compare with strict price stability. Then show that no Taylor rule $i = \bar{\imath} + \psi_y(y - y_e) + \psi_\pi(p - p^e)$ with a constant intercept implements the optimum under productivity shocks: use $a = 0.004$ and $a = -0.004$.

Solution:

c) (4 points) Pessimism: long-run productivity is expected to fall, $\bar{a} = -0.05$, with $a = 0$. Compute $\bar{y}_n$ and r_n. Show that the zero bound binds and compute output and prices at $i = 0$ with $\bar{p} = p^e$. What $\bar{p}$ restores the optimum? Give the limits of the outcome as $\kappa \to \infty$ and $\kappa \to 0$, and discuss credibility.

Solution:

d) (4 points) With borrowers and savers (share χ of savers, initial debt d_0), Benigno's AD is

$$y = \bar{y}_n - \varphi[i - (\bar{p}_n - p) - \rho] + \tfrac{1}{\chi}(g - \bar{g}) + \tfrac{1-\chi}{\chi}[\hat{d} + d_0(p - p^e)], \qquad \varphi \equiv \frac{\sigma\chi + (1-\chi)d_0\beta}{\chi},$$

where $\hat{d}$ is the change in the debt limit. (i) Derive dy/dp when the long-run price $\bar{p}_n$ is anchored and when inflation $\bar{p}_n - p$ is constant, and name the two channels. (ii) Derive the spending multipliers in both regimes. (iii) With $\chi = 2/3$, $d_0 = 1.2$, $\beta = 0.98$, compute them for $\alpha = 0.75$ and $\alpha = 0.66$. (iv) Compute the output response to $\hat{d} = -0.1$ and explain the paradox of flexibility.

Solution: